A CASE-STUDY BASED REPORT

ON

Product Demand Forecasting Using Ridge and Lasso Regression

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INDEX

Abstract

Introduction.....

Background.....
Motivation for the Project .....
Overview of the Work Carried Out.....
Scope and Responsible-Use Statement .....

Problem Statement .....

Case-Study Scenario .....
Central Case-Study Question.....
Formal Problem Definition .....
Objectives that Guide the Study .....
Assumptions and Limitations .....

Dataset and Exploratory Data Analysis

About the Dataset .....
Purpose of EDA .....
Preparation Steps Actually Performed.....
EDA Exploration - Implementation .....
Summary of EDA Findings .....

Algorithm Selection and Implementation.....

Conclusion

Signature of the Student

Abstract

This case-study report presents a supervised machine-learning approach for forecasting product order demand in a retail setting. The project uses historical retail demand data containing product, warehouse, category, date, promotion, holiday, and petrol-price information. The target variable is Order_Demand, making the task a continuous regression problem. The workflow includes data cleaning, exploratory data analysis, date-based feature extraction, outlier treatment, categorical encoding, target encoding of Product_Code, feature scaling, and a 70/15/15 train-validation-test split. An unregularized Linear Regression model is used as a baseline and is compared with Ridge (L2) and Lasso (L1) regression. Ridge and Lasso are tuned using 5-fold cross-validation over 20 alpha candidates. The best Ridge alpha is approximately 26.4 and the best Lasso alpha is approximately 0.34. The three models show very similar validation performance, with RMSE around 9,298 and R^2 around 0.333. Although regularization does not substantially improve raw predictive accuracy in this dataset, Lasso provides useful feature selection by setting 14 of 45 engineered feature coefficients to zero. The analysis identifies product identity through target encoding, warehouse, and selected product-category features as major demand drivers.

Keywords: retail demand forecasting, supervised learning, regression, Ridge regression, Lasso regression, regularization, feature selection, exploratory data analysis.

1. Introduction

1.1 Background

Retail businesses need reliable estimates of product demand to support inventory planning, procurement, and operational decisions. Demand is affected by several factors at the same time, including the product itself, warehouse, category, promotions, holidays, calendar effects, and external variables. A machine-learning regression model can learn relationships between these inputs and historical order demand and then estimate demand for new observations.

1.2 Motivation for the Project

The dataset contains multiple categorical variables, particularly Warehouse and Product_Category. One-hot encoding these variables produces many related features, creating a setting where ordinary linear regression can become less stable and less interpretable. This motivates the use of regularized regression. Ridge reduces the impact of large coefficients through an L2 penalty, while Lasso uses an L1 penalty and can set less useful coefficients exactly to zero.

1.3 Overview of the Work Carried Out

- Collected and examined the Retail_Dataset.csv retail demand data.
- Checked missing values and duplicate records and found none in the checks performed.
- Parsed the date field and extracted Year, Month, Day, and DayOfWeek.
- Explored demand distribution, warehouse-level demand, numerical correlations, and product-category demand.
- Capped Order_Demand at the 99th percentile to reduce the influence of extreme bulk-order values.
- Dropped Product_id because it had 169,211 unique values across 169,211 records and therefore acted as a row-level identifier.
- Created Is_Weekend and Promo_Or_Holiday and target-encoded Product_Code using training data only.
- Split the records into 70% training, 15% validation, and 15% test sets before fitting target encoding and scaling.
- Trained Linear Regression, Ridge, and Lasso and tuned Ridge/Lasso using 5-fold cross-validation.
- Compared models using RMSE, MAE, MAPE, and R^2.

Project repository: github.com/Kairav09/ML-Project

1.4 Scope and Responsible-Use Statement

The scope of the project is limited to forecasting Order_Demand from the variables available in the supplied retail dataset. The model is intended as a decision-support tool for demand estimation and should not be treated as a guaranteed forecast. Because the model is trained on historical observations, changes in market conditions, product availability, pricing strategy, or unusual events may reduce its reliability. For real operational deployment, predictions should therefore be combined with current business information and human judgment.

2. Problem Statement

2.1 Case-Study Scenario

Consider a retailer that records daily product orders across multiple warehouses and product categories. The retailer wants to estimate the expected order demand for a product on a given day so that inventory and procurement can be planned more effectively. The available historical data contains product identifiers, warehouse and category information, dates, promotion and holiday indicators, and petrol-price information.

2.2 Central Case-Study Question

Can a supervised regression model learn useful demand patterns from the available retail features, and can Ridge and Lasso regularization provide a more stable and interpretable solution when categorical features create a relatively large feature space?

2.3 Formal Problem Definition

Let X represent the engineered input features and y represent Order_Demand. The task is to learn a function f(X) that estimates y for unseen observations. The project evaluates an unregularized linear regression baseline and two regularized linear regression models. Ridge minimizes prediction error with an L2 coefficient penalty, while Lasso minimizes prediction error with an L1 coefficient penalty. The regularization strength is controlled by alpha.

2.4 Objectives that Guide the Study

- Build a reproducible preprocessing and feature-engineering pipeline for retail demand data.
- Understand the main demand patterns through exploratory data analysis.
- Reduce the influence of extreme bulk-order observations without deleting potentially valid records.
- Represent categorical information in a form suitable for regression.
- Compare Linear Regression, Ridge Regression, and Lasso Regression.
- Use cross-validation to select suitable regularization strengths.
- Identify influential features and evaluate whether Lasso provides useful automatic feature selection.
- Assess model performance on held-out data using regression metrics.

2.5 Assumptions and Limitations

- Historical relationships in the dataset are assumed to contain useful information about future demand.
- The 99th-percentile cap is assumed to reduce distortion from extreme orders while retaining the main demand pattern.
- Product_Code target encoding is fitted using training data only to avoid target leakage.
- Product_id is treated as a row-level identifier rather than a predictive numerical variable.

- The project does not include richer lag and rolling-window demand features, which could improve short-term forecasting.
- The models are linear and may not capture all nonlinear relationships in retail demand.
- The observed R^2 of about 0.33 indicates that substantial demand variation remains unexplained by the current feature set.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

The project uses Retail_Dataset.csv, a historical retail demand dataset sourced from Kaggle. It contains 169,211 records, four warehouses, 30 product categories, and 2,117 unique products.

Column / Feature	Role or Description
Product_Code	Product identifier used for target encoding.
Product_id	Row-level identifier; dropped during feature engineering.
Warehouse	Warehouse category; four categories: A, C, J, and S.
Product_Category	Product category; 30 categories.
Date	Daily date of the observation.
Order_Demand	Target variable representing product order demand.
Open	Store/warehouse status indicator.
Promo	Promotion indicator.
StateHoliday	State holiday indicator.
SchoolHoliday	School holiday indicator.
Petrol_price	Daily petrol-price index.

3.2 Purpose of EDA

Exploratory data analysis was used to understand the target distribution, identify unusual observations, inspect differences between warehouses and product categories, and determine which groups of features were likely to provide useful predictive information. The EDA also guided subsequent feature-engineering decisions.

3.3 Preparation Steps Actually Performed

- Missing-value checks were performed; no missing values were found in the examined dataset.
- Duplicate checks included full-row duplicates and duplicate Product-Date combinations; no such duplicates were found.
- Date values were parsed using the M/D/YYYY format and transformed into Year, Month, Day, and DayOfWeek.
- Order_Demand was capped at the 99th percentile. This is a Winsorization-style treatment: extreme values are capped rather than removed.
- Warehouse, StateHoliday, and Product_Category were one-hot encoded.
- Petrol_price was standardized using StandardScaler.
- Product_id was removed because each of the 169,211 records had a distinct Product_id.

- Is_Weekend and Promo_Or_Holiday were engineered from the calendar and promotion/holiday information.
- Product_Code was target-encoded using average Order_Demand, with the mapping fitted on training data only.
- Year, Month, Day, DayOfWeek, Petrol_price, and Product_Code target encoding were scaled using training-set statistics.

3.4 EDA Exploration - Implementation

The demand distribution was examined before and after outlier treatment. The original Order_Demand distribution was extremely right-skewed, with rare bulk orders reaching approximately 2.5 million units. Because these observations may represent real business activity, the project capped them rather than deleting the corresponding records.

Warehouse-level demand was also compared. Whse_S and Whse_C showed higher median demand than Whse_A, indicating that warehouse identity can provide useful predictive information. The numerical correlation analysis showed weak individual linear relationships with demand, suggesting that categorical variables could carry important information.

Product-category analysis showed that Category_033 had average demand more than twice that of the next highest category. The uneven category-level demand supported keeping Product_Category in the model and provided motivation for using Lasso feature selection.

3.5 Summary of EDA Findings

- Order_Demand is highly right-skewed and contains rare extreme bulk orders.
- Warehouse is associated with noticeable differences in demand levels.
- Individual raw numerical features have weak linear correlations with demand.
- Product categories have strongly uneven demand patterns.
- The EDA supports the use of categorical features and motivates regularization when many encoded columns are present.

4. Algorithm Selection and Implementation

4.1 Train-Validation-Test Strategy

The dataset was divided into 70% training, 15% validation, and 15% test records. The training set contains approximately 118,000 records. The training portion is the largest because the model needs enough observations to learn stable relationships. The validation set is used during model development and tuning, while the test set is kept separate for final evaluation.

4.2 Baseline: Linear Regression

Ordinary Least Squares Linear Regression was used as an unregularized baseline. It provides a reference point for determining whether adding Ridge or Lasso regularization changes predictive performance.

4.3 Ridge Regression (L2)

Ridge Regression adds a penalty proportional to the sum of squared coefficient values. This discourages very large coefficients and helps distribute the effect among correlated features. Ridge generally shrinks coefficients toward zero but does not normally make them exactly zero. In the project, RidgeCV with 5-fold cross-validation selected a best alpha of approximately 26.4.

4.4 Lasso Regression (L1)

Lasso Regression adds a penalty proportional to the sum of absolute coefficient values. Unlike Ridge, Lasso can reduce some coefficients exactly to zero. This makes Lasso useful for automatic feature selection, especially when the feature set contains many one-hot encoded category columns. LassoCV with 5-fold cross-validation selected a best alpha of approximately 0.34.

4.5 Hyperparameter Tuning with 5-Fold Cross-Validation

Twenty alpha candidates were generated using a log-spaced search from 0.001 to 1000. For each candidate, 5-fold cross-validation repeatedly trained the model on four folds of the training data and validated it on the remaining fold. Every fold therefore served as a validation fold once. The cross-validation results were used by RidgeCV and LassoCV to select the alpha that performed best across the folds.

Item	Project Setting
Baseline	Linear Regression (OLS), unregularized
Ridge	RidgeCV, 5-fold cross-validation
Lasso	LassoCV, 5-fold cross-validation, max_iter=5000
Alpha candidates	20, generated with np.logspace(-3, 3, 20)
Best Ridge alpha	Approximately 26.4
Best Lasso alpha	Approximately 0.34
Training records	Approximately 118,000 (70% of dataset)

Library	scikit-learn

4.6 Model Evaluation

The project evaluates regression performance using RMSE, MAE, MAPE, and R^2. The main validation comparison showed that OLS, Ridge, and Lasso performed almost identically, with RMSE around 9,298 and R^2 around 0.333. This indicates that, for this dataset size and feature set, regularization was not primarily needed to overcome severe overfitting.

Even though the predictive scores were similar, Lasso provided an additional interpretability benefit. In the final engineered feature space, Lasso set 14 of 45 feature coefficients to zero. The strongest signal came from Product_Code target encoding, followed by Warehouse and selected Product_Category dummy variables. Promo, SchoolHoliday, Is_Weekend, and Promo_Or_Holiday contributed smaller but non-zero effects.

4.7 Final Model Choice

If one model must be selected for this project, Lasso is preferred over Ridge because it provides similar predictive performance while also performing automatic feature selection. This makes the final model easier to interpret and helps separate the more informative category features from lower-impact features.

4.8 Reference Research Paper and Relation to the Project

The reference paper, “Retail Demand Forecasting: A Comparative Study for Multivariate Time Series” by Haque, Amin, and Miah (2023), studies retail demand forecasting using Lasso, Ridge, LightGBM, XGBoost, and Decision Tree models. It evaluates demand forecasting with and without macroeconomic variables such as Consumer Price Index, Consumer Sentiment, and unemployment rates. The paper therefore relates directly to this project through the shared retail-demand forecasting problem and its use of Ridge and Lasso regression.

The main difference is the focus. The reference paper emphasizes the contribution of macroeconomic variables and compares a wider range of algorithms, whereas this project focuses on the effect of categorical retail features, regularized linear regression, and Lasso-based feature selection. The paper also reports model performance mainly through RMSE and MAE, while this project additionally uses R^2 and other regression metrics.

5. Conclusion

This case study developed a supervised regression workflow for predicting retail product order demand using historical data. The project combined data cleaning, exploratory analysis, outlier treatment, categorical encoding, target encoding, scaling, and a 70/15/15 data split with three regression models: Linear Regression, Ridge, and Lasso.

The analysis showed that demand is strongly right-skewed, warehouse and product category provide meaningful information, and the raw numerical variables alone have relatively weak linear relationships with demand. The feature-engineering stage therefore retained the categorical information while removing the row-level Product_id and adding calendar and promotion/holiday features.

The model results showed that OLS, Ridge, and Lasso achieved approximately the same validation performance, with RMSE around 9,298 and R^2 around 0.333. Therefore, regularization did not produce a major accuracy improvement for this particular dataset. However, Lasso was more useful from an interpretability perspective because it automatically removed 14 of 45 engineered features by setting their coefficients to zero.

Hence, if one model has to be selected from Ridge and Lasso for this project, Lasso is preferred because it provides comparable predictive performance while also performing feature selection. The strongest demand signal came from the target-encoded product identity, followed by warehouse and selected product categories.

Future Scope

- Add lag and rolling-window demand features to capture recent demand momentum and temporal patterns.
- Compare the linear models with nonlinear models such as Random Forest and XGBoost.
- Investigate whether different product categories benefit from different regularization strengths.
- Build a Streamlit or FastAPI interface around the trained Lasso model for practical use.

Reference

Haque, M. S., Amin, M. S., & Miah, J. (2023). Retail Demand Forecasting: A Comparative Study for Multivariate Time Series.

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